

Developement Over Time (H6)

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```
# Load Packages  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.1      v stringr   1.5.2  
## v ggplot2    4.0.0      v tibble    3.3.0  
## v lubridate  1.9.4      v tidyr     1.3.1  
## v purrr      1.1.0  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)
```

```
## Loading required package: Matrix  
##  
## Attaching package: 'Matrix'  
##  
## The following objects are masked from 'package:tidyr':  
##  
##     expand, pack, unpack
```

```
library(lmerTest)
```

```
##  
## Attaching package: 'lmerTest'  
##  
## The following object is masked from 'package:lme4':  
##  
##     lmer  
##  
## The following object is masked from 'package:stats':  
##  
##     step
```

```

# Load Interleaf data
df <- read.csv("../Data/df_interleaf_measures.csv")

# Create datasets for human-human and human-ai conditions
df_hh <- df %>%
  filter(shared_condition_order == "without_ai")

df_hai <- df %>%
  filter(shared_condition_order == "with_ai")

```

Flow

```

# human-human
# Null Model
flow_hh_null <- lmer(flow_rating ~ shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hh,
  REML = FALSE)

```

```

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00204693 (tol = 0.002, component 1)

```

```

# Full Model
flow_hh_full <- lmer(flow_rating ~ chat_number + shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hh,
  REML = FALSE)

```

```

# Compare Models
anova(flow_hh_null, flow_hh_full)

```

```

## Data: df_hh
## Models:
## flow_hh_null: flow_rating ~ shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
## flow_hh_full: flow_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID) +
##              npar      AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## flow_hh_null    7 1542.1 1563.9 -764.03    1528.1
## flow_hh_full    8 1523.4 1548.3 -753.68    1507.4 20.699  1  5.374e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

summary(flow_hh_full)

```

```

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: flow_rating ~ chat_number + shared_prompt_order + (1 + chat_number |
##          participant_ID) + (1 | dyad_ID)

```

```
## Data: df_hh
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    1523.4    1548.4    -753.7    1507.4      160
##
## Scaled residuals:
##      Min      1Q   Median      3Q      Max
## -2.49729 -0.50374  0.02123  0.55750  1.97708
##
## Random effects:
##   Groups             Name             Variance Std.Dev. Corr
## participant_ID (Intercept) 320.303   17.897
##               chat_number    2.791    1.671   -0.76
## dyad_ID       (Intercept)  78.315    8.850
## Residual                        312.796   17.686
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    36.29257    5.33068   61.13405    6.808 4.90e-09 ***
## chat_number      3.57162    0.67609   42.31115    5.283 4.15e-06 ***
## shared_prompt_order 0.02168    1.25938  125.82236    0.017  0.986
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cht_nm
## chat_number -0.556
## shrd_prmpt_ -0.513 -0.135
```

```
# human-AI
# Null Model
flow_hai_null <- lmer(flow_rating ~ shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hai,
  REML = FALSE)

# Full Model
flow_hai_full <- lmer(flow_rating ~ chat_number + shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hai,
  REML = FALSE)

# Compare Models
anova(flow_hai_null, flow_hai_full)
```

```
## Data: df_hai
## Models:
## flow_hai_null: flow_rating ~ shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
## flow_hai_full: flow_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
##              npar      AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## flow_hai_null    7 1498.1 1519.9 -742.03   1484.1
```

```
## flow_hai_full      8 1488.4 1513.4 -736.20    1472.4 11.661  1  0.0006382 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(flow_hai_full)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: flow_rating ~ chat_number + shared_prompt_order + (1 + chat_number |
## participant_ID) + (1 | dyad_ID)
## Data: df_hai
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  1488.4   1513.4   -736.2    1472.4      160
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.44029 -0.53045  0.04655  0.54931  2.09041
##
## Random effects:
## Groups          Name          Variance Std.Dev. Corr
## participant_ID (Intercept) 550.714  23.467
##                chat_number   3.728   1.931  -0.45
## dyad_ID        (Intercept) 100.031  10.002
## Residual                184.537  13.584
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    41.5310     5.4195  42.8154   7.663 1.47e-09 ***
## chat_number      2.1521     0.5896  37.6341   3.650 0.000793 ***
## shared_prompt_order 2.0146     0.9914 125.9945   2.032 0.044257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cht_nm
## chat_number -0.482
## shrd_prmpt_ -0.415 -0.082
```

Social presence

```
# human-human
# Null Model
social_presence_hh_null <- lmer(social_presence_rating ~ shared_prompt_order +
                               (1 + chat_number | participant_ID) +
                               (1 | dyad_ID),
                               data = df_hh,
                               REML = FALSE)

# Full Model
```

```
social_presence_hh_full <- lmer(social_presence_rating ~ chat_number + shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hh,
  REML = FALSE)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00208486 (tol = 0.002, component 1)
```

```
# Compare Models
```

```
anova(social_presence_hh_null, social_presence_hh_full)
```

```
## Data: df_hh
```

```
## Models:
```

```
## social_presence_hh_null: social_presence_rating ~ shared_prompt_order + (1 + chat_number | participant_ID)
```

```
## social_presence_hh_full: social_presence_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID)
```

```
##
```

	npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df
social_presence_hh_null	7	1528.7	1550.6	-757.35	1514.7		
social_presence_hh_full	8	1519.8	1544.8	-751.91	1503.8	10.87	1

```
##
```

```
## Pr(>Chisq)
```

```
## social_presence_hh_null
```

```
## social_presence_hh_full 0.0009774 ***
```

```
## ---
```

```
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(social_presence_hh_full)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
```

```
## method [lmerModLmerTest]
```

```
## Formula: social_presence_rating ~ chat_number + shared_prompt_order +
```

```
## (1 + chat_number | participant_ID) + (1 | dyad_ID)
```

```
## Data: df_hh
```

```
##
```

```
##
```

	AIC	BIC	logLik	-2*log(L)	df.resid
	1519.8	1544.8	-751.9	1503.8	160

```
##
```

```
## Scaled residuals:
```

```
##
```

	Min	1Q	Median	3Q	Max
	-2.54476	-0.44007	0.06066	0.57783	2.23911

```
##
```

```
## Random effects:
```

```
##
```

Groups	Name	Variance	Std.Dev.	Corr
participant_ID	(Intercept)	4.316e+02	20.77432	
	chat_number	1.194e+01	3.45510	-0.77
dyad_ID	(Intercept)	4.251e-05	0.00652	
Residual		2.856e+02	16.89893	

```
##
```

```
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
```

```
##
```

```
## Fixed effects:
```

```
##
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	51.5317	5.1762	67.6750	9.956	6.9e-15 ***
chat_number	2.8885	0.8087	46.6704	3.572	0.000835 ***

```
##
```

```
## shared_prompt_order -0.2988      1.2649 117.2150 -0.236 0.813701
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) cht_nm
## chat_number -0.639
## shrd_prmpt_ -0.528 -0.116
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00208486 (tol = 0.002, component 1)
```

```
# human-AI
# Null Model
social_presence_hai_null <- lmer(social_presence_rating ~ shared_prompt_order +
                                (1 + chat_number | participant_ID) +
                                (1 | dyad_ID),
                                data = df_hai,
                                REML = FALSE)

# Full Model
social_presence_hai_full <- lmer(social_presence_rating ~ chat_number + shared_prompt_order +
                                (1 + chat_number | participant_ID) +
                                (1 | dyad_ID),
                                data = df_hai,
                                REML = FALSE)

# Compare Models
anova(social_presence_hai_null, social_presence_hai_full)
```

```
## Data: df_hai
## Models:
## social_presence_hai_null: social_presence_rating ~ shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
## social_presence_hai_full: social_presence_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
##          npar    AIC    BIC  logLik -2*log(L)  Chisq Df
## social_presence_hai_null      7 1516.9 1538.8 -751.46    1502.9
## social_presence_hai_full      8 1517.8 1542.8 -750.91    1501.8 1.1122  1
##          Pr(>Chisq)
## social_presence_hai_null
## social_presence_hai_full      0.2916
```

```
summary(social_presence_hai_full)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: social_presence_rating ~ chat_number + shared_prompt_order +
##          (1 + chat_number | participant_ID) + (1 | dyad_ID)
## Data: df_hai
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##      1517.8      1542.8      -750.9      1501.8        160
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -3.2256 -0.3396 0.0555 0.4487 3.5119
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## participant_ID (Intercept) 616.88 24.837
## chat_number 2.08 1.442 -0.25
## dyad_ID (Intercept) 75.60 8.695
## Residual 229.48 15.149
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 57.7227 5.6805 29.2379 10.162 4.19e-11 ***
## chat_number 0.6465 0.6030 37.6181 1.072 0.2905
## shared_prompt_order 2.2383 1.0774 125.8078 2.078 0.0398 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cht_nm
## chat_number -0.442
## shrd_prmpt_ -0.437 -0.075
```

Affect

```
# human-human
# Null Model
affect_hh_null <- lmer(panas_rating ~ shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hh,
  REML = FALSE)

# Full Model
affect_hh_full <- lmer(panas_rating ~ chat_number + shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hh,
  REML = FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Compare Models
anova(affect_hh_null, affect_hh_full)
```

```
## Data: df_hh
## Models:
## affect_hh_null: panas_rating ~ shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
## affect_hh_full: panas_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
##          npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
```

```
## affect_hh_null      7 1413.5 1435.3 -699.73    1399.5
## affect_hh_full      8 1409.8 1434.8 -696.90    1393.8 5.6615  1    0.01734 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(affect_hh_full)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: panas_rating ~ chat_number + shared_prompt_order + (1 + chat_number |
## participant_ID) + (1 | dyad_ID)
## Data: df_hh
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    1409.8    1434.8    -696.9    1393.8      160
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.01635 -0.44738 -0.01616  0.50132  2.34645
##
## Random effects:
## Groups          Name          Variance Std.Dev. Corr
## participant_ID (Intercept)    6.281   2.506
##                chat_number    2.298   1.516   1.00
## dyad_ID        (Intercept) 100.777  10.039
## Residual                148.815  12.199
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    57.1253    3.6212   58.3261  15.775 <2e-16 ***
## chat_number     1.2051    0.4931   51.1703   2.444  0.018 *
## shared_prompt_order 0.3200    0.8750  128.1176   0.366  0.715
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cht_nm
## chat_number -0.340
## shrd_prmpt_ -0.525 -0.130
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# human-AI
# Null Model
affect_hai_null <- lmer(panas_rating ~ shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hai,
  REML = FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Full Model
affect_hai_full <- lmer(panas_rating ~ chat_number + shared_prompt_order +
  (1 + chat_number | participant_ID) +
  (1 | dyad_ID),
  data = df_hai,
  REML = FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Compare Models
anova(affect_hai_null, affect_hai_full)
```

```
## Data: df_hai
## Models:
## affect_hai_null: panas_rating ~ shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
## affect_hai_full: panas_rating ~ chat_number + shared_prompt_order + (1 + chat_number | participant_ID) + (1 | dyad_ID)
##
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
affect_hai_null	7	1387.5	1409.4	-686.75	1373.5			
affect_hai_full	8	1385.7	1410.7	-684.84	1369.7	3.8109	1	0.05092

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(affect_hai_full)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: panas_rating ~ chat_number + shared_prompt_order + (1 + chat_number |
## participant_ID) + (1 | dyad_ID)
## Data: df_hai
##
```

	AIC	BIC	logLik	-2*log(L)	df.resid
	1385.7	1410.7	-684.8	1369.7	160

```
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.77143 -0.49448  0.05298  0.57714  2.97690
##
## Random effects:
## Groups           Name                Variance Std.Dev. Corr
## participant_ID (Intercept) 2.035e+02 14.26649
## chat_number      chat_number 1.122e-03  0.03349 -1.00
## dyad_ID          (Intercept) 1.245e+02 11.15757
## Residual                    1.092e+02 10.45080
## Number of obs: 168, groups: participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	56.9225	4.1561	44.5974	13.696	<2e-16 ***
chat_number	0.7540	0.3825	127.7280	1.971	0.0508 .
shared_prompt_order	1.1225	0.7229	126.0033	1.553	0.1230

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##      (Intr) cht_nm
## chat_number -0.395
## shrd_prmpt_ -0.406 -0.068
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```